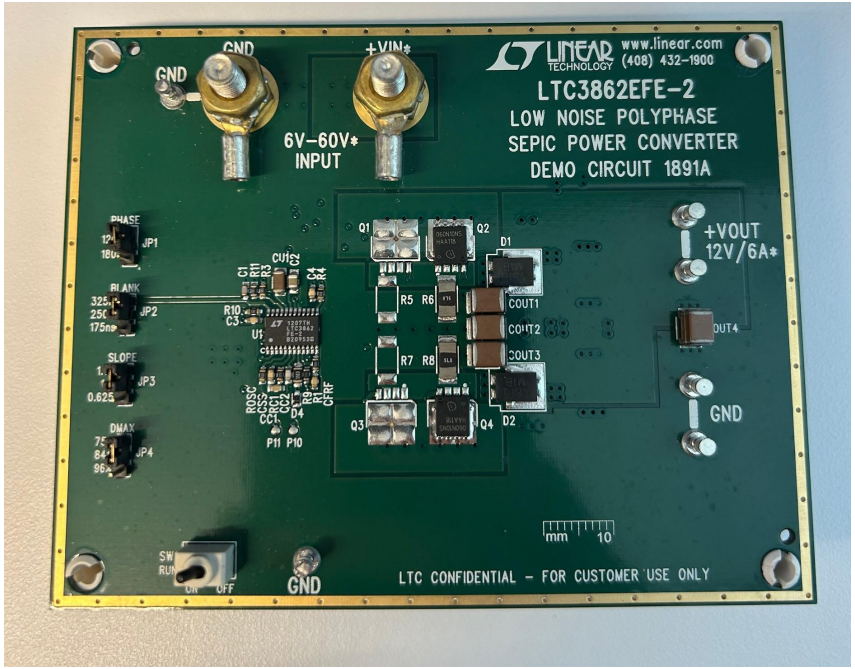
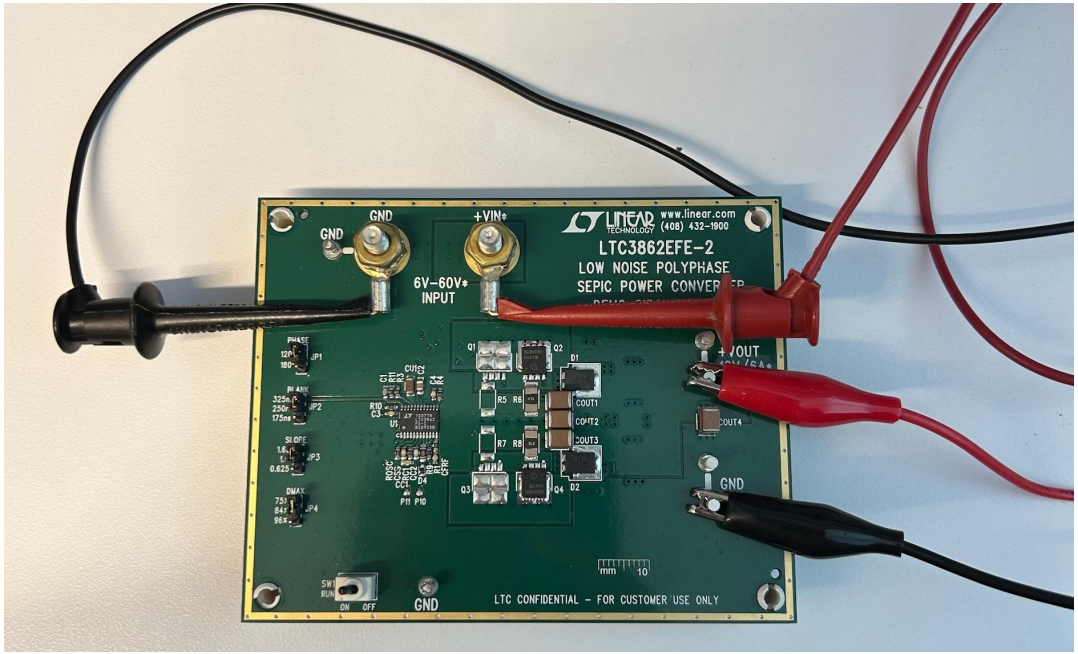
PHASE 4

- Get the evaluation board.
- Set up the lab instrumentation for measurements.
- Complete measurements on board and take snapshots of measurements, waveforms etc.
- Make sure to look at output ripple measurements.
- Make slides of your experimental work:
  - Measurement setup and results, scope waveforms.
  - Make sure to look at ripple measurements

SEPIC Converter PCB



SEPIC Converter PCB - Hooked Up



10V input

1 5.00V/ 2 3 4 0.0s 5.00s/ Auto? 1 12.0V
Agilent
Acquisition
Normal
4.00GSa/s
Channels
DC 1.00:1
DC 1.00:1
DC 1.00:1
DC 1.00:1
Measurements
DC RMS - FS(1): 11.9V
Pk-Pk(1): 0.0V
Measure Current Mean Min Max Std Dev Count
DC RMS - FS(1): 11.9V 11.895V 11.8V 11.9V 22.276mV 8.289k
Pk-Pk(1): 0.0V 94.788mV 0.0V 500mV 131.61mV 8.289k
Measurement Menu
Source 1
Type: Pk-Pk
Add Measurement
Settings
Clear Meas
Statistics

20V input

1 5.00V/ 2 3 4
0.0s 5.00s/ Auto?
1 12.0V
Agilent
Acquisition
Normal
4.00GSa/s
Channels
DC 1.00:1
DC 1.00:1
DC 1.00:1
DC 1.00:1
Measurements
DC RMS - FS(1): 11.9V
Pk-Pk(1): 0.0V
Measure Current Mean Min Max Std Dev Count
DC RMS - FS(1): 11.9V 11.893V 11.8V 11.9V 25.614mV 8.756k
Pk-Pk(1): 0.0V 96.597mV 0.0V 500mV 132.16mV 8.756k
Measurement Menu
Source 1
Type: Pk-Pk
Add Measurement
Settings
Clear Meas
Statistics

30V Input

1 5.00V/ 2 3 4 0.0s 5.00s/ Auto? 1 12.0V
Agilent
Acquisition
Normal
4.00GSa/s
Channels
DC 1.00:1
DC 1.00:1
DC 1.00:1
DC 1.00:1
Measurements
DC RMS - FS(1): 11.9V
Pk-Pk(1): 0.0V
Measure Current Mean Min Max Std Dev Count
DC RMS - FS(1): 11.9V 11.858V 2.3V 11.9V 507.24mV 151.6k
Pk-Pk(1): 0.0V 100.30mV 0.0V 500mV 133.54mV 151.6k
Measurement Menu
Source 1
Type: Pk-Pk
Add Measurement
Settings
Clear Meas
Statistics